Kinematika mobilnih robota

Branimir Ćaran
Konzultacije: Četvrtkom po dogovoru
branimir.caran@fsb.unizg.hr

Projektiranje autonomnih sustava| Regional Center of Excellence for Robotic Technology

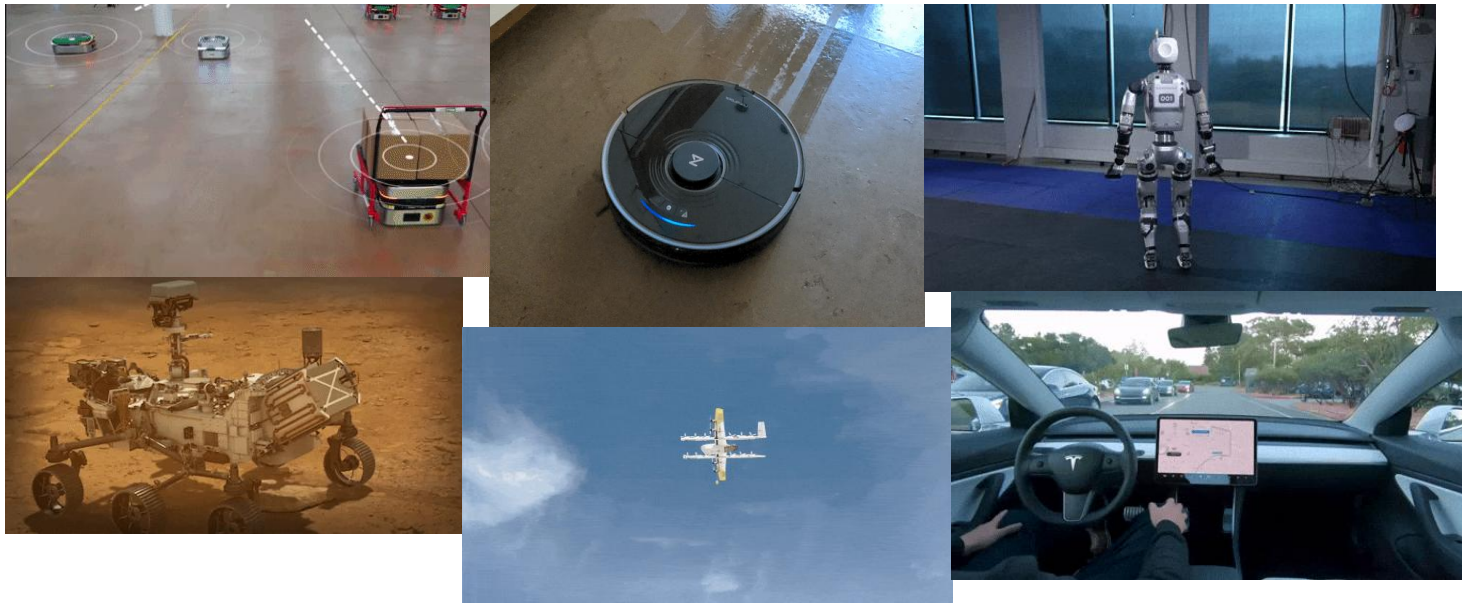
Sadržaj

- Mobilni roboti
- Podjela mobilnih robota
- Kinematski modeli
- Dijagram toka
- Primjeri

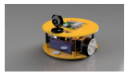
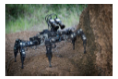
crta

Mobilni robot

- Mobilni robot je robot koji je sposoban ostvariti gibanje u prostoru pomoću kotača, gusjenica, nogu ili propelera
- Primjeri u praksi:
  - Industrija (logistika, AGV u tvornicama)
  - Dostava (roboti za dostavu)
  - Usluge (Roomba, dostava hrane)
  - Istraživanje (roveri na Marsu)
  - Autonomna vozila



- Ključna pitanja kod projektiranja autonomnih mobilnih sustava:
  - Kako se robot smije/mora gibati u prostoru? (holonomni ili neholonomni)
  - Kako povezati gibanje kotača s gibanjem robota? (direktna kinematika)
  - Kako iz željenog gibanja dobiti naredbe za kotače? (inverzna kinematika)

Domain	Type	Subtype	Image	Description
Air-Based	Fixed wing	Classical Fixed Wing		Traditional fixed-wing robots for long-range and efficient forward flight, [35]
		VTOL		Vertical Takeoff and Landing fixed-wing aircraft, combining hover and forward flight, [42]
		VTOL Tilt-copter		Fixed-wing aircraft with tilting rotors for vertical lift and transition to forward flight, [39]
	N-copters	–		Multi-rotor robots for vertical takeoff, hovering, and agile maneuverability, [14]
Ground-Based	Tracked	–		Robots with continuous caterpillar tracks for excellent rough terrain traversal, [23]
	Wheeled	Ackermann		Car-like steering mechanism for ground robots, [44]
		Differential Drive		Independent control of left and right wheels for turning, [41]
		Omniwheeled		Specialized wheels allowing movement in any direction, [36]
		Mecanum		Diagonal rollers enable sideways and diagonal movement, [32]
	Legged	Bipedal		Two-legged robots mimicking human walking motion, [21]
		Quadrupedal		Four-legged robots for stability, agility, and uneven terrain, [15]
		Hexapod		Six-legged robots for redundancy and exceptional rough terrain handling, [2]
Marine	Subs	–		Underwater robots used for exploration, inspection, and research missions, [1]

Trajectory optimization for autonomous overtaking manuevers – M.R.Cihlar (FSB 2025.)

Kinematika robota

- Holonomnost sustava:
  - **Neholonomni** sustavi imaju kinematska ograničenja koja im onemogućuju trenutno kretanje u svakom smjeru ravnine
  - **Holonomni** sustavi imaju mogućnost trenutnog kretanja u bilo kojem smjeru
- Kinematika robota:
  - **Manipulator** - položaj efektora računa se iz **trenutnih vrijednosti zglobova** pomoću matrica transformacija.

$$ {}^{0}T_n = {}^{0}T_1 \, {}^{1}T_2 \ldots {}^{n-1}T_n $$

  - **Mobilni roboti** – položaj robota računa se **integracijom brzina robota** kroz vrijeme

$$ x(t) = \int \dot{x}(t)dt \, , y(t) = \int \dot{y}(t)dt \, , \theta(t) = \int \dot{\theta}(t)dt $$

  - Direktna kinematika – iz poznatih brzina kotača robota računa se brzina cijelog robota
  - Inverzna kinematika – iz željenih brzina robota računaju se brzine kotača kako bi se to ostvarilo

Model monocikla

- *Engl. Unicycle model,* neholonoman robot
- x_b, y_b - to je koordinatni sustav vezan uz tijelo robota, smješten u točki kontakta kotača s tlom
- x_w, y_w - globalni koordinatni sustav
- x, y – koordinate robota u globalnom koordinatnom sustavu
- Robotu zadajemo željene brzine:

$v = \begin{bmatrix} v_{b,x} \\ \omega_{b,z} \end{bmatrix}$, $v_{b,x}$ - linearna brzina robota u smjeru osi robota x_b

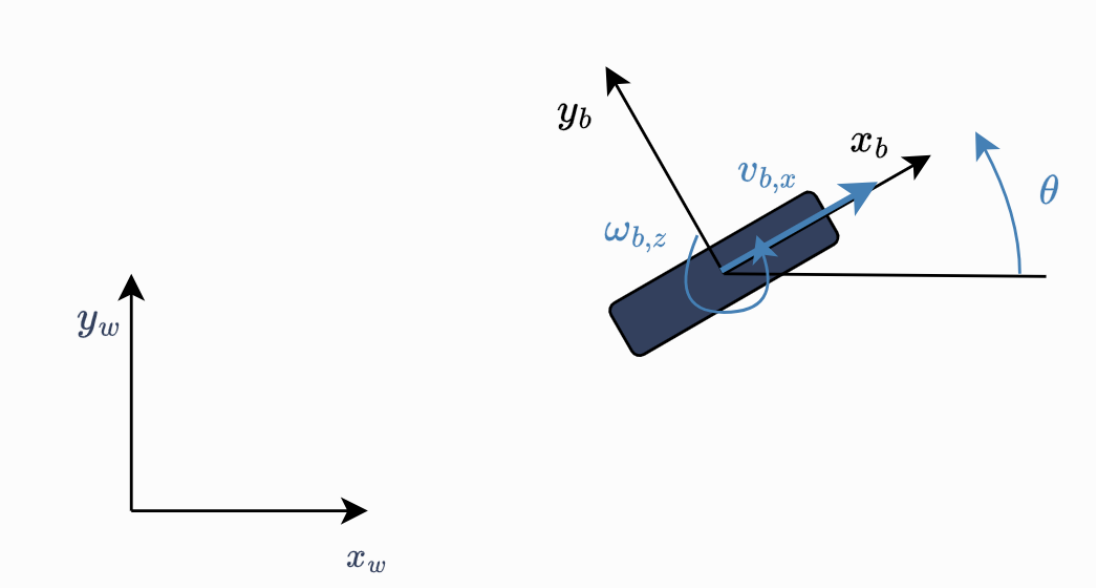
$\omega_{b,z}$- kutna brzina robota oko osi robota z_b

Direktna kinematika:

$$x = v_{b,x} \cos(\theta)$$

$$y = v_{b,x} \sin(\theta)$$

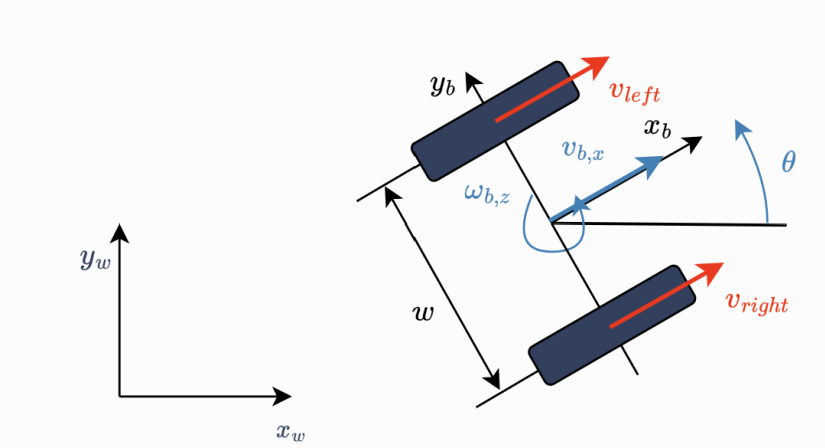
$$\dot{\theta} = \omega_{b,z}$$



A unicycle in the strict sense (i.e., a vehicle equipped with a single wheel) is a robot with a serious problem of balance in static conditions. However, there exist vehicles that are kinematically equivalent to a unicycle but more stable from a mechanical viewpoint. Siciliano et.al - Robotics: Modelling, Planning and Control:

Model diferencijalnog robota

- *Engl. Differential Drive Robot,* neholonoman robot
- w- razmak između kotača (engl. *wheel track*)
- v_{left}, v_{right} - linearne brzine lijevog i desnog kotača



Direktna kinematika:

$$v_x = \frac{v_{right} + v_{left}}{2}$$

$$\omega_{b,z} = \frac{v_{right} - v_{left}}{w}$$

Položaj u globalnom koordinatnom sustav:

$$\dot{x} = v_{b,x}\cos(\theta)$$

$$\dot{y} = v_{b,x}\sin(\theta)$$

$$\dot{\theta} = \omega_{b,z}$$

Inverzna kinematika:

$$v_{left} = v_{b,x} - \omega_{b,z}\frac{w}{2}$$

$$v_{right} = v_{b,x} + \omega_{b,z}\frac{w}{2}$$

Model bicikla

- *Engl. Bicycle,* neholonoman robot
- l- razmak između kotača *(engl. wheelbase)*
- v_{front}, v_{rear} - linearne brzine prednjeg i zadnjeg kotača
- ϕ – kut skretanja prednjeg kotača

Uvjet bez proklizavanja:

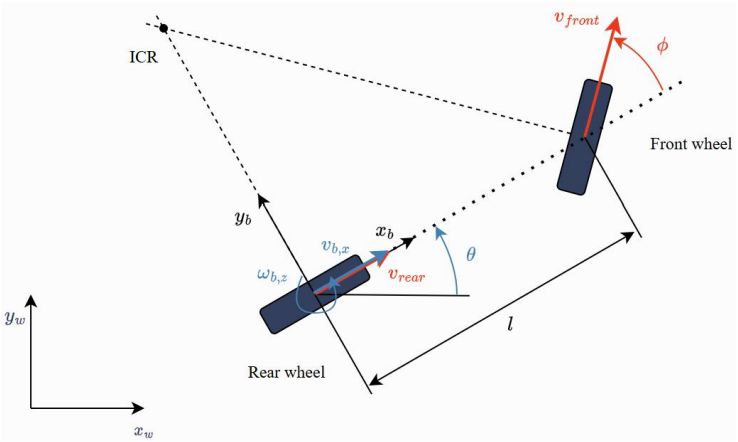
$$v_{rear} = v_{front}\cos(\phi)$$

Direktna kinematika

$$\dot{x} = v_{b,x}\cos(\theta)$$

$$\dot{y} = v_{b,x}\sin(\theta)$$

$$\dot{\theta} = \frac{v_{b,x}}{l}\tan(\phi)$$



Inverzna kinematika:

$$v_{rear} = v_{b,x}$$

$$\phi = \mathrm{atan}\left(\frac{l\omega_{b,z}}{v_{b,x}}\right)$$

$$v_{front} = \frac{v_{b,x}}{\cos(\phi)}$$

Pretpostavljamo da se kotači kotrljaju bez proklizavanja. To znači da je brzina točke kontakta kotača s tlom jednaka nuli, a brzina kotača usmjerena je u smjeru okomitom na os kotača. **Trenutačni centar rotacije (ICR)**, tj. središte kružnice oko koje se robot rotira, nalazi se u presjeku pravaca koji su okomiti na osi kotača i prolaze kroz točke kontakta kotača s tlom.

Ackermann model

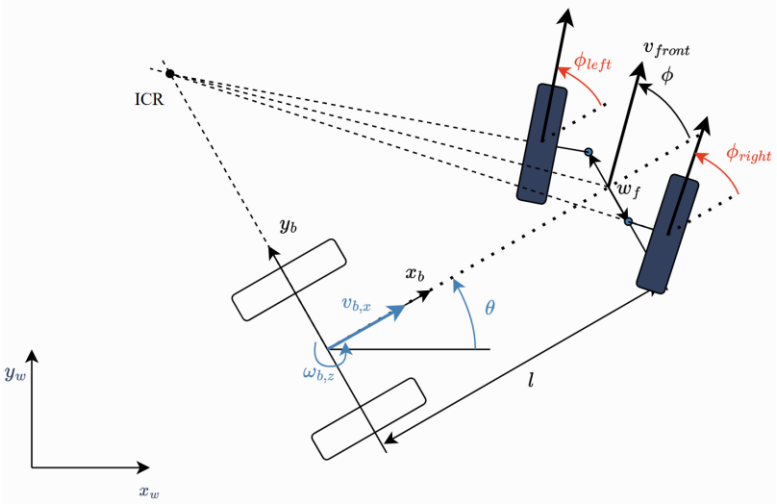
- *Engl. Ackermann steering,* neholonoman robot
- w_f - udaljnost kotača prednje osovine, mjerena između dvaju zglobova skretanja *(engl. kingpin)*
- Kako bi se spriječilo proklizavanje prednjih kotača, njihovi kutovi skretanja ne mogu biti jednaki - Ackermannovo upravljanje.

Direktna kinematika (ista kao za model bicikla)

$$\dot{x} = v_{b,x}\cos(\theta)$$

$$\dot{y} = v_{b,x}\sin(\theta)$$

$$\dot{\theta} = \frac{v_{b,x}}{l}\tan(\phi)$$



Inverzna kinematika:

$$R_b = \frac{l}{\tan(\phi)}$$

$$\phi_{left} = \operatorname{atan}\left(\frac{l}{R_b - w_f/2}\right) = \operatorname{atan}\left(\frac{2l\sin(\phi)}{2l\cos(\phi) - w_f\sin(\phi)}\right)$$

$$\phi_{right} = \operatorname{atan}\left(\frac{l}{R_b + w_f/2}\right) = \operatorname{atan}\left(\frac{2l\sin(\phi)}{2l\cos(\phi) + w_f\sin(\phi)}\right)$$

$$\phi = \operatorname{atan}\left(\frac{l\tan(\phi_{left})}{l + w_f/2\tan(\phi_{left})}\right) = \operatorname{atan}\left(\frac{l\tan(\phi_{right})}{l - w_f/2\tan(\phi_{right})}\right)$$

Ackermannovo upravljanje može se ostvariti i mehaničkom vezom između dva prednja kotača. U tom slučaju robot ima samo jedan upravljački ulaz, a upravljački kutovi dvaju prednjih kotača mehanički su povezani

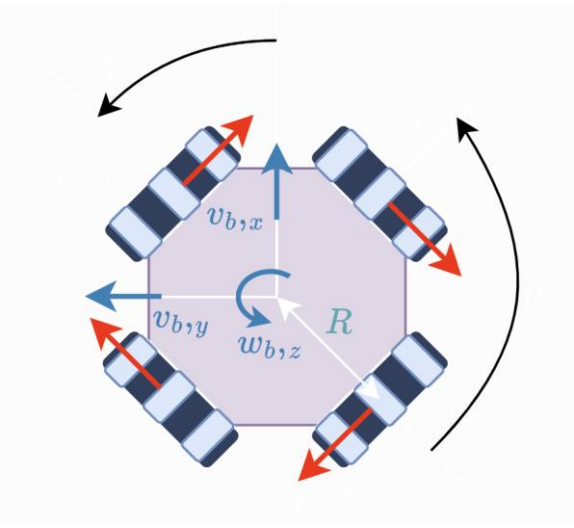
Svesmjerni robot

- *Engl. Omnidirectional Drive Robots,* holonoman robot
- R- radijus robota, udaljenost kotača od centra robota
- Crvena strelica označava pozitivan smjer svakog kotača
- γ-označava kut prvog kotača od osi robota x_b
- δ- označava kut između svakog kotača i može se izračunati s obzirom na ukupan broj kotača n

$$\delta = \frac{2\pi}{n}$$

- Direktna kinematika može se izračunati koristeći pseudo inverz matrice **A**:

$$\begin{bmatrix} v_{b,x} \\ v_{b,y} \\ \omega_{b,z} \end{bmatrix} = \mathbf{A}^{\dagger} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$



Inverzna kinematika:

$$\mathbf{A} = \begin{bmatrix} \sin(\gamma) & -\cos(\gamma) & -R \\ \sin(\delta+\gamma) & -\cos(\delta+\gamma) & -R \\ \vdots & \vdots & \vdots \\ \sin((n-1)\delta+\gamma) & -\cos((n-1)\delta+\gamma) & -R \end{bmatrix}$$

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = \mathbf{A} \begin{bmatrix} v_{b,x} \\ v_{b,y} \\ \omega_{b,z} \end{bmatrix}$$

$$v_i = \sin((n-1)\delta+\gamma) * v_{b,x} - \cos\big((n-1)\delta+\gamma\big)\, v_{b,y} - R\omega_{b,z}$$

Swerve drive robot (4WIS4WID)

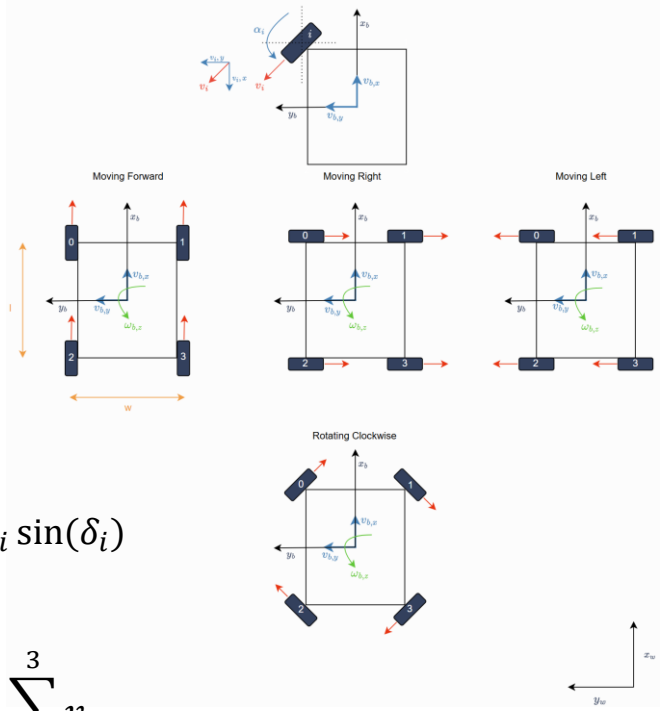
- Holonoman robot
- Lokacije kotača $(l_{i,x}, l_{i,y})$ u odnosu na bazu robota su definirane kao:
  - Prednji lijevi kotač: $\left(\frac{l}{2}, \frac{w}{2}\right)$
  - Prednji desni kotač: $(\frac{l}{2}, -\frac{w}{2})$
  - Stražnji lijevi kotač: $(-\frac{l}{2}, \frac{w}{2})$
  - Stražnji lijevi kotač: $(-\frac{l}{2}, -\frac{w}{2})$
- Inverzna kinematika:
- Za svaki kotač definirane su komponente brzina:

$$\begin{bmatrix} v_{i,x} \\ v_{i,y} \end{bmatrix} = \begin{bmatrix} v_{b,x} - \omega_{b,z} \cdot l_{i,y} \\ v_{b,y} + \omega_{b,z} \cdot l_{i,x} \end{bmatrix}$$

- Potom se za svaki kotač računa linearna brzina i kut zakreta:

$$v_i = \sqrt{v_{i,x}^2 + v_{i,y}^2}$$

$$\delta_i = \operatorname{atan2}(v_{i,y}, v_{i,x})$$



Direktna kinematika:

$$v_{i,x} = v_i \cos(\delta_i)\,, v_{i,y} = v_i \sin(\delta_i)$$

Brzine robota su potom računaju kao:

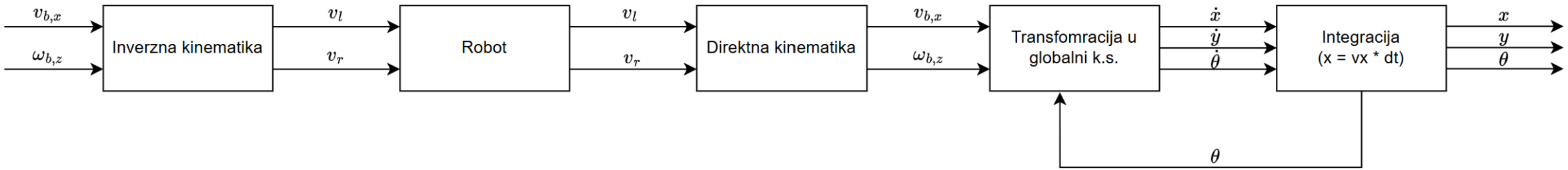
$$v_{b,x} = \frac{1}{4}\sum_{i=0}^{3} v_{i,x}\ , v_{b,y} = \frac{1}{4}\sum_{i=0}^{3} v_{i,y}$$

$$\omega_{b,z} = \frac{\sum_{i=0}^{3}(v_{i,y} l_{i,x} - v_{i,x} l_{i,y})}{\sum_{i=0}^{3}(l_{i,x}^2 + l_{i,y}^2)}$$

Dijagram toka

- Primjer za diferencijalnu strukturu
- Ulazne brzine može davati čovjek (preko joystick-a) ili neki nadređeni regulator za slijeđenje putanje ili trajektorije
- Robot preko enkodera motora očitava impulse koje pretvara u brzinu svakog motora (rotacijsku ili translacijsku):

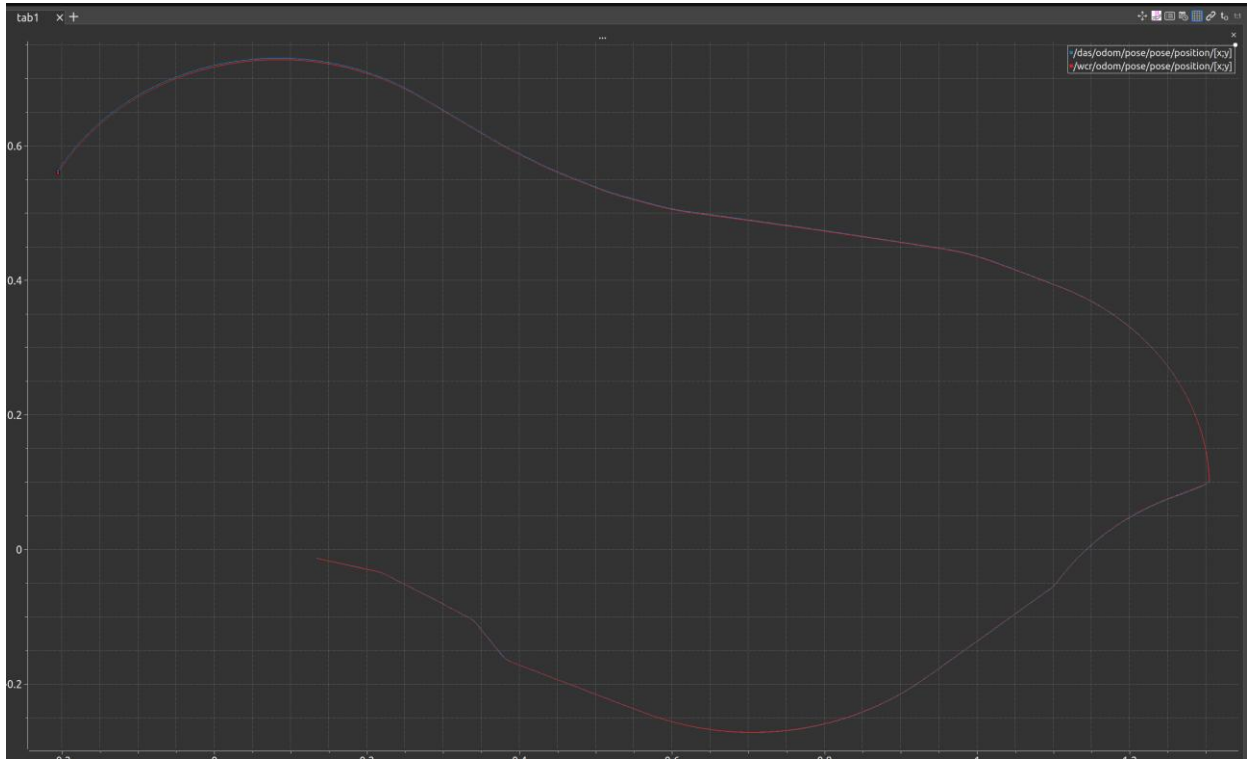
$$\omega_m = \frac{n[k] - n[k-1]}{N} \cdot \frac{2\pi}{t[k] - t[k-1]} \left\langle \frac{rad}{s} \right\rangle \rightarrow v = \omega_m \cdot r$$



Zadatak za vježbu

```
git clone https://github.com/CRTA-Lab/das_odometry_examples.git
```

- Potrebno je preuzeti pakete s *github*-a
- Unutar repozitorija se nalaze 2 paketa, jedan s gotovom 4WIS4WID odometrijom napisan u C++
- Drugi paket sadrži template za Python unutar kojeg je potrebno napisati kod koji će iz */joint_states* topica računati i objavljivati odometriju
- Možete koristiti *wcr/odom* topic kako bi provjerili vlastiti čvor za odometriju



Hvala!

Branimir Ćaran

Konzultacije: Četvrtkom 12:00-13:00
branimir.caran@fsb.unizg.hr